Daniel Raj S

Agentic Quality Engineer | AI-Driven Test Automation | SDET

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PROFESSIONAL SUMMARY

AI-focused Quality Engineer with 7+ years in test automation, building agentic quality systems across web, mobile, and API. Designs AI workflows on Claude and AWS Bedrock for failure triage, accessibility remediation, and quality intelligence, backed by Playwright/TypeScript, Selenium/Java, Appium, Rest Assured, and CI/CD expertise. Scrum lead reducing flaky tests and time-to-resolution while delivering actionable quality metrics.

PROFESSIONAL EXPERIENCE

Coursera | Test Engineer 2 (IC3) | Jul 2025 – Present

- Sole QA engineer and quality gatekeeper for the Integration POD, leading the Infrastructure team's shift from manual QA to automated E2E quality for third-party LMS/LXP and business APIs across Coursera for Campus (C4C) and Coursera for Business (C4B).
- Designed and built an AI-powered QE Intelligence Dashboard surfacing per-POD quality scores, team performance, SLO burn rates, error budgets, and bug-resolution metrics for real-time release decisions.
- Engineered an agentic Playwright trace-analysis pipeline (Claude Skill, Jenkins, AWS Bedrock) that triages failures, returns an RCA verdict, and opens fix PRs for flaky tests, cutting on-call TTR from 2 hours to 20 minutes.
- Built an agentic a11y remediation pipeline that renders the affected component on a real page, validates fixes with Claude on Bedrock, and generates Playwright specs for component and full-page checks pre-PR.
- Developed Playwright CLI plugins for test-case/test-plan generation and maintain 12+ CI/CD pipelines, reducing manual test-design effort by 30%, pipeline failures by 40%, and deployment time by 35%.

Tegna (Premion), Chennai | Senior Automation Engineer | Jul 2024 – Jul 2025

- Served as Scrum Lead for the automation team, facilitating sprint planning, daily stand-ups, reviews, and retrospectives while championing BDD across QA, engineering, and business stakeholders.
- Built a hybrid automation framework from the ground up with Playwright, TypeScript, and Cucumber.js, raising UI/API regression coverage from 45% to 85% and cutting regression cycle time from 2 days to 6 hours.
- Embedded SonarQube static analysis into CI/CD, reducing critical code-quality violations by 30% pre-merge.

Bank of New York Mellon, Chennai | QA Analyst | Jun 2022 – Apr 2024

- Designed and implemented a Java Selenium automation framework (JUnit, TestNG, Maven) with a reusable POM and Apache POI, integrated with GitLab Runner CI/CD to reduce manual testing effort by 25%.
- Supported a 20-member QA team as one of 3 automation engineers; led migration of 150+ manual regression cases to automation, cutting manual testing workload by 40%.

Tata Consultancy Services (TCS), Chennai | QA Engineer I | Dec 2019 – May 2022

- Automated critical test cases in Java with Selenium WebDriver and TestNG, achieving 70% coverage; built a reusable POM framework and adapted test strategy across Agile sprints.

TECHNICAL SKILLS

Programming Languages & Data: Java, Python, SQL, TypeScript, PostgreSQL, Snowflake, AWS Athena
Test Automation: Playwright, Selenium, Cucumber.js/Java, Appium, Rest Assured, Apache POI, a11y automation
AI & Agentic Engineering: Claude Code/Skills, AWS Bedrock, agentic workflows, AI RCA, Playwright MCP
Quality Analytics: SLO burn rates, error budgets, quality scoring, bug-resolution metrics, QE dashboards
CI/CD & DevOps: Jenkins, GitLab CI/CD, GitHub Actions, Docker, AWS (EC2, S3), SonarQube, Node.js
API & Other Testing: Postman, SoapUI, Swagger, JMeter | **Collaboration:** Jira, Confluence, Scrum

ACHIEVEMENTS

- Reduced defect-escape rate by 30% by integrating automated smoke and regression suites into CI/CD.
- Won "Best Automation Hack" (Octillion 2024) for a CI plugin surfacing test coverage metrics on dashboards.

EDUCATION

Alagappa University | Master of Computer Applications | 2021 – 2023
Madurai Kamaraj University | Bachelor of Computer Application | 2016 – 2019